

VR STRESS RESPONSE TRAINER



Capstone Project Phase B

Project Number 26-1-D-18

Sapir Gerstman | Ido Ben Amara

Advisor: Dr. Moshe Sulamy

GITHUB



BACKGROUND

Emergency situations demand fast decisions under extreme psychological pressure. Civilians and first responders often lack realistic, safe training that also measures how stress affects their performance.

SOLUTION

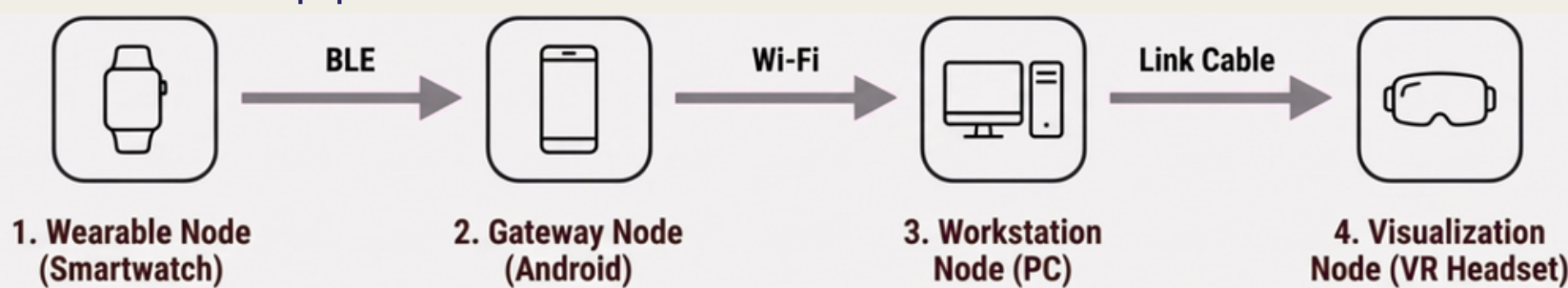
VR Stress Response Trainer is an immersive VR training platform that combines:

- Realistic emergency scenarios.
- Live physiological monitoring (Heart Rate & HRV).
- Personalized stress feedback via the Stress Change Index (SCI).

ARCHITECTURE & TECHNOLOGIES

- Samsung smartwatch - Samples HR & HRV (PPG sensor).
- Android application - Relays data to PC.
- Unity 3D - Main platform (Game logic, Simulations, Analytics).
- Meta Quest headset - Immersive VR training environment.

Distributed pipeline across four nodes:



SIMULATIONS

- **Environment Learning:** Explore key locations with no alarm.
- **Simulation 1:** The trainee must respond to the alarm by turning off lights, securing the entrance, collecting essential equipment and reaching the shelter as quickly and calmly as possible.
- **Simulation 2:** The trainee must use the first aid kit, assess and treat the injured person, then reach the public phone and call emergency services (101) to report the incident.

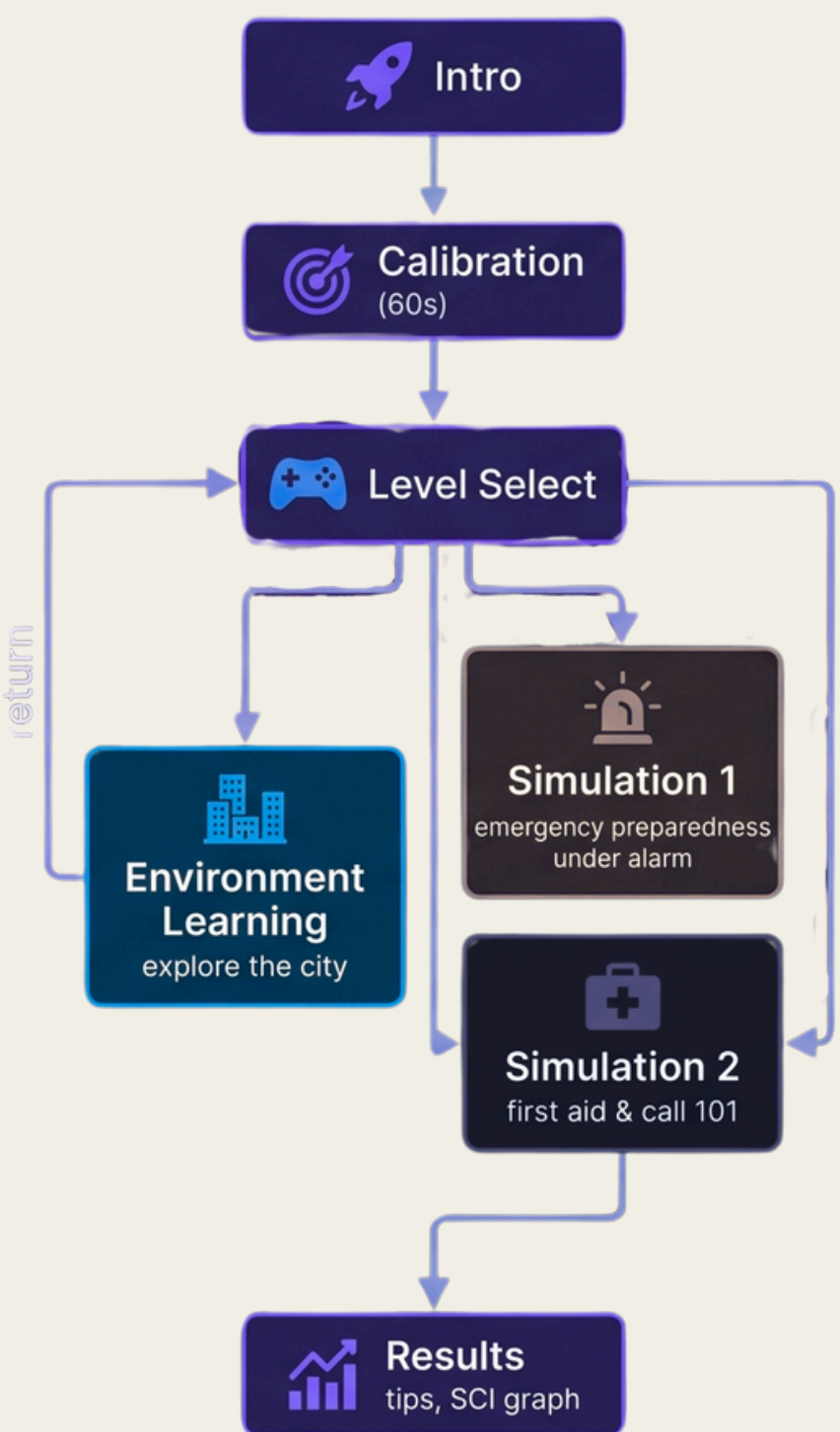
GOAL

Improve self-regulation, situational awareness and decision-making under pressure in a controlled environment.

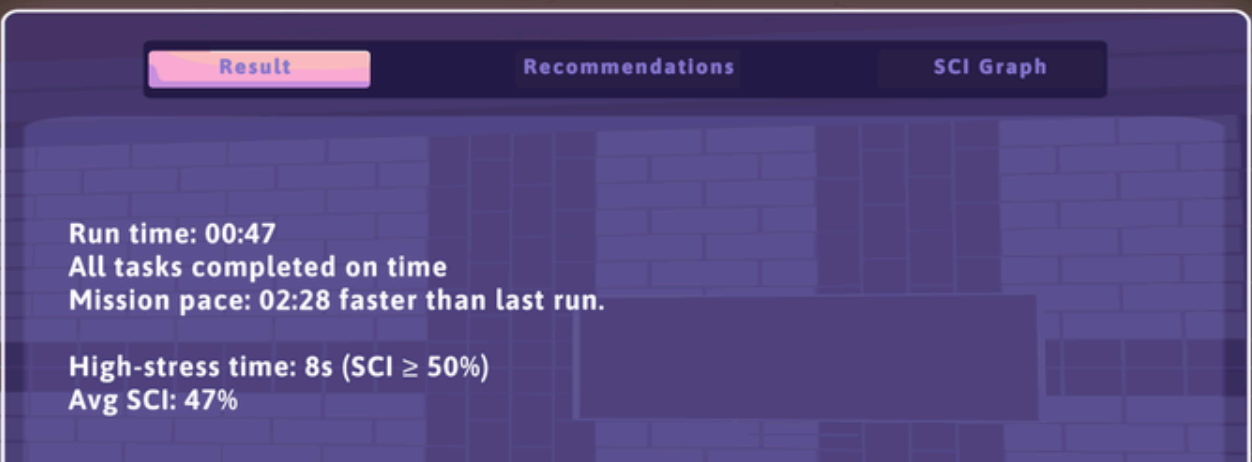
KEY REQUIREMENTS

- 60s baseline HRV calibration.
- Two emergency simulations (home safety, first aid).
- Real-time HR/HRV streaming from smartwatch.
- SCI-based stress .
- Post-session metrics & personalized recommendations.
- User profile & session history.

MAIN USER FLOW



Results Simulation 1



Results Simulation 1

